

Raghav Senthil Kumar

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, BuildIllinois, Innovation LLC, Agentic AI

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Built a SaaS platform on *AWS* to automate research & case-writing in competitive high school debate, reaching **10K+ users**
- Designed a text retrieval algorithm that extracts evidence segments from a debate case using URL validation & IQR filtering. Indexed **600K+ documents** on an *Elasticsearch* engine, returning query results in **less than 1 second**
- Developed a de-duplication algorithm that flagged **30%** of evidence documents as identical, reducing index bloat

Product Development Intern | Jetson

January 2025 – February 2025

- Conducted **7 iterations of UI/UX testing** across 3 experimental features on the Jetson App (100K+ users)
- Delivered a **5-page report** on recommending strategies to optimize user engagement directly to the CEO
- Redesigned user onboarding on *Figma*, cutting it from **12 to 5 pages**, validated via **A/B testing** on 30 test users

Data Analytics Intern | STEM E

June 2024 – July 2024

- Analyzed 3 years of TikTok data (**200+ posts**) on *Hootsuite* and *Microsoft Excel* to find patterns in outreach
- Identified that “science fun facts” posts comprised 69.3% of views, despite accounting for 7% of all content
- Pitched new content strategy centered around “fun facts” that **increased engagement by 24% (8K+ views)**

PROJECTS

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Led team of 4 to build a VLM-based pricing algorithm from user-inputted truck images for **Kamion (YC S22)**
- Constructed dataset of **15,000 images** scraped from publicly-available truck listings and audited for data quality
- Wrote *JSON schema* for **Claude Sonnet 4.6 VLM** to extract features (condition, brand, model, year) that define the pricing algorithm and filter out non-truck images. Fine-tuned across **30 independent iterations of RLHF**
- Optimized pricing algorithm via *quantile regression* with **87% accuracy**. **2nd at UIUC Founder’s Hackathon**

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Engineered a **suite of 3 edge AI tools** that collects data regarding a school’s energy, water and food consumption
- Built a *Python FastAPI* backend that ingests live water, waste & energy data from all 3 tools, runs **4 Random Forest models** to detect anomalies & deliver actionable steps via **Gemma Voice LLM** on a *Next.js* dashboard
- Led a team of 5 across project ideation, development and presentation. **Finalist at USAII Global Hackathon**

EcoBin | [Research Paper](#) | [Source Code](#)

March 2026 – May 2026

- Designed a two-stage *EfficientNetV2-S* neural network (**4.39M** parameters) trained on **15K+ images** that classifies waste into 4 disposal pathways, achieving **96.13% overall accuracy**. Published 2 original datasets onto *Kaggle*
- Built a synthetic data generation pipeline using *U2-Net* segmentation and alpha-compositing to generate 9K images to train the model to detect contamination in recyclable waste. **Placed 2nd at the Arizona State Science Fair**
- Engineered a smart bin prototype using *Arduino Uno* that sorts the waste item with **less than 2 second latency**

Chronos | [Live Website](#) | [Source Code](#)

December 2025

- Built a **scheduling agent** (*React/Firebase*) that auto-books meetings for large groups (30+ ppl) on Google Calendar
- Designed a **5-step Python scheduling algorithm** that scans calendar availability, returns a ranked list of time slots and sends email confirmation for each booking. Validated algorithm across **14 unit and integration test cases**
- Created a lobby system that enables the host to authenticate invited participants and queue follow-up meetings
- Licensed Chronos to **8 startups** & signed **2 LOIs worth \$3.5K**. **Won 2025 Congressional App Challenge**

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, SQL, HTML/CSS

Frameworks & Databases: FastAPI, Flask, React, Next.js, Node.js, PostgreSQL, Elasticsearch, Firebase, Supabase

Cloud & Tools: AWS, Google Cloud, Vercel, Docker, Git/GitHub, CI/CD, REST APIs, Unit Testing, Claude Code

AI/ML: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, Hugging Face, pandas, NumPy, Claude API, LLMs, RAG